Which constitutive model do measured cavitation records prefer?

PyIMR

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Abstract

We compare seventeen constitutive models against laser-induced cavitation records in gelatin at three temperatures, using Bayesian model selection with a noise scale measured from trial-to-trial spread and a prior that prices a model down where it merely reproduces a simpler one it contains. A quadratic Zener solid — strain-stiffening elasticity *and* stress relaxation — wins at every temperature, reaching a residual of $\chi^2/N = 1.23$ to 1.35 against experimental repeatability. Neither ingredient suffices alone: relaxation without stiffening reaches 1.41 to 2.94, and every memoryless model lands above 12. The advantage of the stiffening term collapses as the gel warms, which is the clearest physical signal in the comparison. We report the limitations plainly: the reported residuals are ceilings rather than estimates, because the winning model's best-fitting region is partly unintegrable.

1 Setup

Each dataset is a set of repeated laser-induced cavitation events at one temperature, recorded as $R(t)/R_{\max}$. Figure 1 shows the mean trace and the trial-to-trial spread, which is what we use as the measurement noise: it is measured rather than assumed, so a residual of $\chi^2/N \approx 1$ means a model reproduces the record to within experimental repeatability.

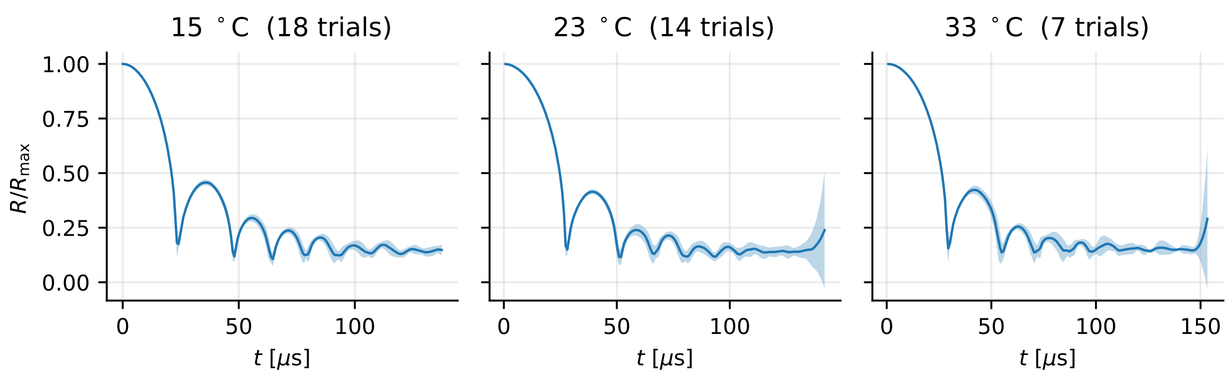

Figure 1: Measured records. Line is the mean over trials, band is ± one standard deviation across trials at each time. The spread is the noise model.

Two samples are excluded before fitting. A trial reading above its own maximum radius is a tracking failure, and the $t = 0$ sample where every trial agrees exactly is a definition rather than a measurement — giving it an error bar would invent an uncertainty and let it constrain the fit.

Each candidate is evaluated on a log-spaced grid over its own parameters, at one resolution for every model so that grid luck cannot decide the winner. The evidence is a grid quadrature over that box, with the noise scale marginalised under a half-Cauchy prior, a redundancy factor that down-weights parameters where a model merely reproduces a simpler model it contains, and a BIC-style complexity penalty. Trajectories are compressible (Keller–Miksis); the incompressible approximation changes the residuals by a factor of two to four and must not be used for these records.

2 What is being computed, and why

Four objects carry the analysis. Each is derived here rather than quoted, because the choices between them are not neutral and two of the errors this study corrected came from treating them as interchangeable.

2.1 The likelihood, and what it assumes

Each record gives a mean radius y_i at time t_i over 18 trials, with the trial-to-trial standard deviation s_i as the noise estimate. Modelling the observations as independent and Gaussian about the prediction $m_i(\theta)$,

$$p(y \mid \theta) = \prod_i \frac{1}{\sqrt{2\pi}\,\sigma_i} \exp\left[-\frac{(y_i - m_i(\theta))^2}{2\sigma_i^2}\right], \qquad -2\log p = \chi^2(\theta) + \sum_i \log\bigl(2\pi\sigma_i^2\bigr). \tag{1}$$

Three assumptions are buried here and each is separately checkable. *Gaussianity* is the weakest claim, and is reasonable because y_i is a mean over 18 trials. *Known* σ_i is not innocuous — using the sample spread as if it were the true deviation understates uncertainty — and is handled in §2.2 by giving the noise scale a prior and integrating it out. *Independence* is the one that fails: the diagonal covariance in (1) says each residual carries fresh information, and §5 measures a lag-one autocorrelation of 0.90.

The deviations are not constant across the record. Optical tracking degrades where the wall moves fastest, so the variance is inflated by a logistic weight in the strain rate, $\sigma_i^2 = s_i^2/w_i$ with $w = w_0 + (1-w_0)\,\mathrm{logit}^{-1}\bigl(\kappa(\dot\varepsilon_{\mathrm{th}} - |\dot\varepsilon_i|)/\dot\varepsilon_{\mathrm{th}}\bigr)$. This is a statement about the measurement, not the model, and it is why χ^2/N is dominated by the late record where the weights are large.

2.2 Integrating out the noise scale

Rather than trust s_i, introduce a single scale β multiplying every deviation, $\sigma_i \to \beta\sigma_i$. Substituting into (1),

$$\log p(y \mid \theta, \beta) = -\frac{\chi^2(\theta)}{2\beta^2} - N\log\beta + \mathrm{const}, \tag{2}$$

which depends on the data only through χ^2 and N. That is what makes the marginalisation cheap: the forward model never re-enters, so

$$\log p(y \mid \theta) = \log \int_0^\infty p(y \mid \theta, \beta)\,\pi(\beta)\,\mathrm{d}\beta \tag{3}$$

costs one pass over a quadrature grid rather than one solve per node. With the half-Cauchy prior $\pi(\beta) = (2/\pi)/(1+\beta^2)$ the integral is done numerically on $\beta \in [0.05, 10]$.

Why marginalise rather than fit β? Profiling at its best value gives $-\frac{1}{2}N\,[1 + \log(\chi^2/N)]$, and marginalising is always smaller. The gap is an Occam penalty: a model that fits only by inflating its own error bars is charged for the inflation. Fitting β would hand that flexibility over for free.

2.3 Why the evidence, and not the best fit

From Bayes' theorem, the posterior odds between two models are the prior odds times the ratio of

$$Z_M = p(y \mid M) = \int p(y \mid \theta, M)\, \pi(\theta \mid M)\, \mathrm{d}\theta. \tag{4}$$

This is the only quantity that answers "which model should I believe" without an arbitrary complexity penalty. Comparing best fits cannot: a nested model can never fit worse than the model containing it, so χ^2 ranks by flexibility. Information criteria add a penalty — $2k$ for AIC, $k \log N$ for BIC — but BIC is an asymptotic approximation to $-2 \log Z$ that assumes a regular, well-identified model with N large, and §2.5 shows this one is not identified in a direction. The integral itself makes no such assumption.

The penalty appears explicitly on expanding (4) about the maximum. With $\hat{\theta}$ the best fit, $H = -\nabla^2 \log p(y \mid \theta)$ evaluated there, and $\theta = \hat{\theta} + \delta$,

$$Z_M \simeq p(y \mid \hat{\theta})\, \pi(\hat{\theta}) \int \exp\Big[-\tfrac{1}{2}\delta^{\mathsf{T}} H \delta\Big]\, \mathrm{d}\delta = \underbrace{p(y \mid \hat{\theta})}_{\text{best fit}}\ \underbrace{\pi(\hat{\theta})\, (2\pi)^{p/2}\, |H|^{-1/2}}_{\text{Occam factor}}. \tag{5}$$

The Gaussian integral is exact for a quadratic $-\log p$, so (5) is the Laplace approximation and its error is governed by the third derivative. Reading the second factor as $\pi(\hat{\theta}) \times$ (posterior volume) makes its meaning plain: it is the fraction of the prior the posterior actually occupies, so a model needing fine tuning pays. This is the sense in which a good fit and a credible model differ, and it is what Freund and Ewoldt argue should replace goodness-of-fit ranking in rheology.

One consequence matters later and is easy to miss. Because $|H|^{-1/2}$ grows as the posterior broadens, a sloppy direction *raises* the evidence. Model comparison therefore cannot detect non-identifiability — it rewards it — and the two questions must be asked separately.

2.4 The prior this study uses

Equation (4) needs $\pi(\theta \mid M)$, and a uniform box would make the answer depend on where the box was drawn. Two factors are used instead, both computed on the same grid the quadrature runs over.

The *bottleneck* maps each parameter to $u_j \in [0, 1]$ in logarithms — the parameters span decades, so a linear map would encode the bounds rather than the physics — and takes the harmonic mean $H(\theta) = p \big/ \sum_j u_j^{-1}$. The harmonic mean is chosen over the arithmetic one precisely because it is dominated by its smallest argument: a parameter driven to the bottom of its range drags the whole prior down, on the reasoning that a parameter the data pushed to a boundary is one the model did not need.

The *redundancy* weight asks whether a model is merely emulating a simpler one it contains. For a candidate and each nested child, the relative stress mismatch F is mapped by $F^n/(F^n + \tau^n)$ against the resolvable scale τ, giving a number near 0 when the parent is imitating the child and approaching 1 when it does something genuinely different. The weight is the *minimum* over children: a model that can imitate any one simpler model is redundant there, however different it looks from the rest.

2.5 The Fisher information, and what its eigenvectors are

Differentiating $-\log p$ twice at fixed data and taking the expectation over noise, the term containing $\partial^2 m$ vanishes because $\mathbb{E}[y_i - m_i] = 0$, leaving

$$F_{jk} = \sum_i \frac{1}{\sigma_i^2} \frac{\partial m_i}{\partial \theta_j} \frac{\partial m_i}{\partial \theta_k} = \left(J^\mathsf{T} J\right)_{jk}, \qquad J_{ij} = \frac{1}{\sigma_i} \frac{\partial m_i}{\partial \theta_j}, \tag{6}$$

so the Fisher information is exactly the Gram matrix of the whitened sensitivities — no approximation, given (1). Near the optimum $\Delta\chi^2 \simeq \Delta\theta^\mathsf{T} F \Delta\theta$, so the surface of constant misfit is the ellipsoid $\Delta\theta^\mathsf{T} F \Delta\theta = c$. Diagonalising $F = V\Lambda V^\mathsf{T}$, its axes are the eigenvectors and their half-lengths are $\sqrt{c/\lambda_k}$: a small eigenvalue is a long axis, a direction along which the data barely object.

The Cramér–Rao bound gives this statistical force. For any unbiased estimator, $\mathrm{Cov}(\hat\theta) \succeq F^{-1}$, so F^{-1} is the best achievable covariance and its eigen-decomposition is the confidence ellipsoid. The standard deviation along direction k is $\lambda_k^{-1/2}$, and the ratio $\kappa = \lambda_{\max}/\lambda_{\min}$ measures how far the problem is from determining all directions equally.

Two conventions are load-bearing. Taking $\theta = \log(\text{parameters})$ makes an eigenvector a power law: moving along v sends $\Delta \log g = v_g t$ and $\Delta \log \alpha = v_\alpha t$, so $g \propto \alpha^{v_g/v_\alpha}$ and the exponent is read directly off the components. And if the sensitivities are taken with respect to a scaled coordinate $u_j = (\log\theta_j - a_j)/w_j$, then $\Delta \log \theta_j = w_j \tilde v_j$ and the exponent is $(w_g \tilde v_g)/(w_\alpha \tilde v_\alpha)$ — with box widths of three and four decades here, a 25% correction, and getting it wrong is what produced a spurious agreement earlier in this work.

Finally, F is the observability Gramian of the system augmented with $\dot\theta = 0$: identifiability and observability are the same statement. The distinction that matters is *structural* unidentifiability, where F is singular and no data can help, against *practical* sloppiness, where F is invertible but ill-conditioned. Here $\lambda_{\min} = 5.2 \times 10^{-2} \neq 0$ and the profile likelihood has a genuine minimum, so the problem is the second — which is why a different experiment can fix it.

2.6 What correlated residuals cost

Equation (1) counts N independent pieces of information. If instead the residuals have autocorrelation ρ_k at lag k, then for a sum of N correlated terms $\mathrm{Var}(\sum r_i) = N\sigma^2\big(1 + 2\sum_k (1 - k/N)\rho_k\big)$, so the information is that of

$$N_{\text{eff}} = \frac{N}{1 + 2\sum_{k\geq 1} \rho_k} \tag{7}$$

independent samples, and parameter uncertainties are understated by $\sqrt{N/N_{\text{eff}}}$. The sum is truncated at the first negative ρ_k — Geyer's initial positive sequence — because the long tail is estimated from few pairs each and contributes sampling error rather than signal.

The correct remedy is a covariance with off-diagonal terms, not a larger σ. Taking $C_{ij} = \sigma_i \sigma_j \exp(-|t_i - t_j|/\tau)$, the Cholesky factor $C = LL^\mathsf{T}$ gives the generalised least squares form

$$-2\log p = r^\mathsf{T} C^{-1} r + \log|C| + \text{const} = \|L^{-1} r\|^2 + 2\sum_k \log L_{kk} + \text{const}, \tag{8}$$

which is what `FieldObservation` computes. Inflating σ instead would treat structure as noise: it widens the error bars while leaving the parameter estimates biased, which is the failure Brynjarsdóttir and O'Hagan warn about.

2.7 Which design criterion, and why they disagree

An experiment is chosen by a scalar summary of F, and the classical family is

$$\text{D: } \max \det F, \qquad \text{A: } \min \operatorname{tr} F^{-1}, \qquad \text{E: } \max \lambda_{\min}, \qquad \text{c: } \min c^{\mathsf{T}} F^{-1} c. \tag{9}$$

D minimises the volume of the confidence ellipsoid, A its mean squared axis length, E its longest axis, and c the variance of one nominated combination. They rank designs differently because they summarise the same spectrum differently, and the choice is a statement about what one wants to learn.

For identifiability the relevant fact is a scaling argument. A design change that only amplifies signal — more samples, less noise, a larger bubble — sends $F \to sF$. Then $\det F \to s^p \det F$ and $\lambda_{\min} \to s\lambda_{\min}$, so D and E both improve, while

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \quad \text{is unchanged.} \tag{10}$$

Degeneracy is a property of the eigen*vectors*, not the eigenvalues, and only a design that rotates them can remove it. This is exactly why the E-optimal search preferred a larger bubble while conditioning preferred a gentler collapse: more signal against better separation.

The c-optimal choice carries a subtler trap. Nominating c as the sloppy direction *of the present design* optimises for a direction that moves once the design changes, and in this study that criterion recommended the opposite of the right answer.

When the model itself is in doubt, none of these apply, since all presuppose the model whose F is being computed. T-optimality asks instead for the design maximising the discrepancy between rival predictions, $\max_\xi \|m_1(\hat\theta_1;\xi) - m_2(\hat\theta_2;\xi)\|$, and privileges neither.

3 Result

Table 1 and Figure 2 give the outcome. The quadratic Zener model `qSLS` — strain-stiffening elasticity with stress relaxation — wins at every temperature, and the ordering beneath it separates cleanly into three tiers: models with both ingredients, models with relaxation only, and memoryless models.

Table 1: Best χ^2/N by model class. Values near 1 indicate a fit within experimental repeatability.

	15 °C	23 °C	33 °C
`qSLS` (stiffening + relaxation)	**1.35**	**1.33**	**1.23**
`SLS` (relaxation, linear elasticity)	2.94	1.85	1.41
viscoelastic fluids	5.3–5.7	4.5–4.7	1.9–2.1
memoryless	12.6–16.7	—	—

Figure 3 overlays the winning model on the data it was selected against.

3.1 Reading the posteriors

The model posteriors saturate: the winner takes essentially all the mass and the runners-up are reported as 10^{-29} or smaller. These magnitudes should not be read literally. Evidence differences scale with sample count, and with roughly 3600 samples a modest per-sample difference in fit

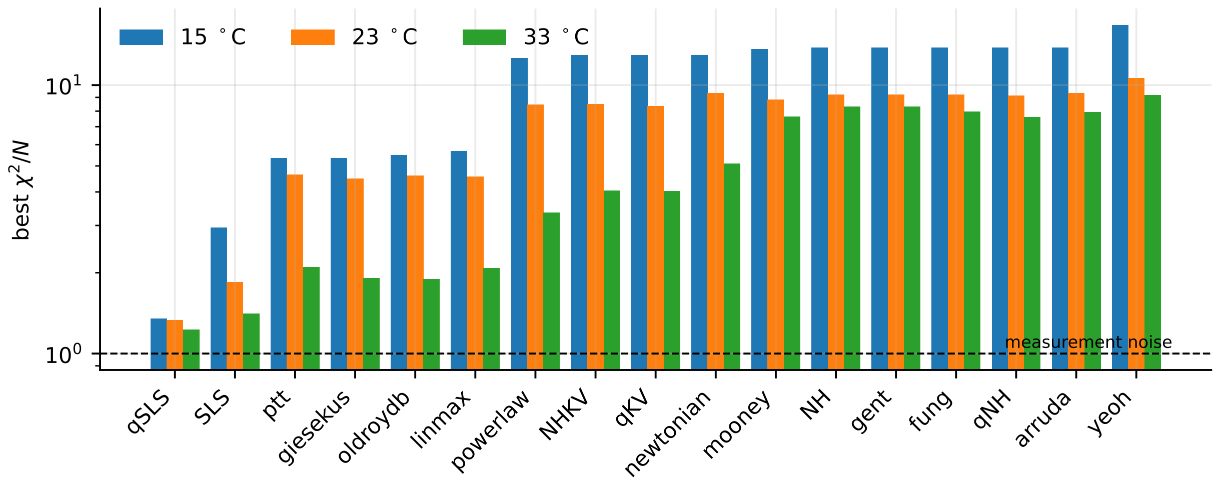

Figure 2: Best χ^2/N for every candidate, all three temperatures, logarithmic scale. The dashed line is the measurement-noise floor.

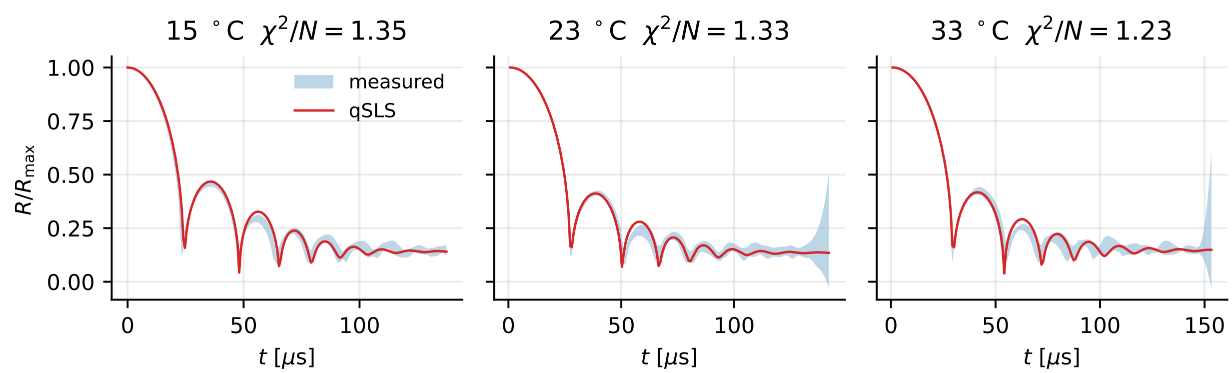

Figure 3: The selected model against the measured band it was fitted to.

compounds into an astronomical Bayes factor. The defensible reading is ordinal — the winner is decisive and the ordering is meaningful, the magnitudes are not. The residual χ^2/N is the quantity to quote.

4 The temperature trend

The most informative result is not any single number but a trend (Figure 4). The advantage that the stiffening term buys over pure relaxation shrinks monotonically as the gel warms: the gap between `SLS` and `qSLS` falls from 2.94 versus 1.35 at 15 °C to 1.41 versus 1.23 at 33 °C. A warmer, softer gel needs less strain-stiffening to explain its record. Because this is a trend across three independent datasets rather than a single fitted value, it is harder to attribute to a fitting artefact than any individual χ^2.

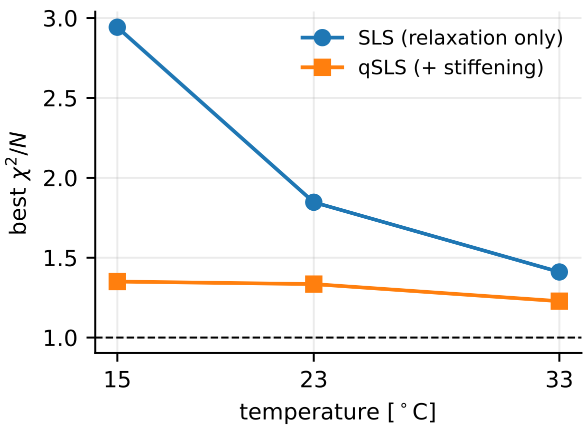

Figure 4: What the stiffening term buys, against temperature. The advantage of `qSLS` over `SLS` narrows as the gel warms.

5 Limitations

The dropped points are outside the model, not outside the solver. Roughly one point in six of the winner's parameter box cannot be integrated. That is not a numerical shortcoming. In that region the model expands to $R/R_0 = 2132$ after its collapse — identically at relative tolerances of 10^{-3}, 10^{-4} and 10^{-5}, so a converged property of the equations rather than an artefact. The bubble begins at its maximum radius and its energy is finite, so growth of that size is the constitutive model failing, and refusing those points is the correct outcome.

The count of physically admissible points is pinned near 1958 of 2401 sampled whatever the settings: looser tolerances and a $20\times$ step budget convert failures into runaways, never into usable solves. This corrects an earlier reading of the same failures as a stiffness problem. It also means the dropped points cannot be recovered by numerical effort, and that no claim about the winner's fit should be made in that region.

The ranking survives; the parameter estimates do not. Independent sequential Monte Carlo runs give the winner 2173.55 ± 0.19 in log evidence against 2071.90 ± 0.38 for the relaxation-only

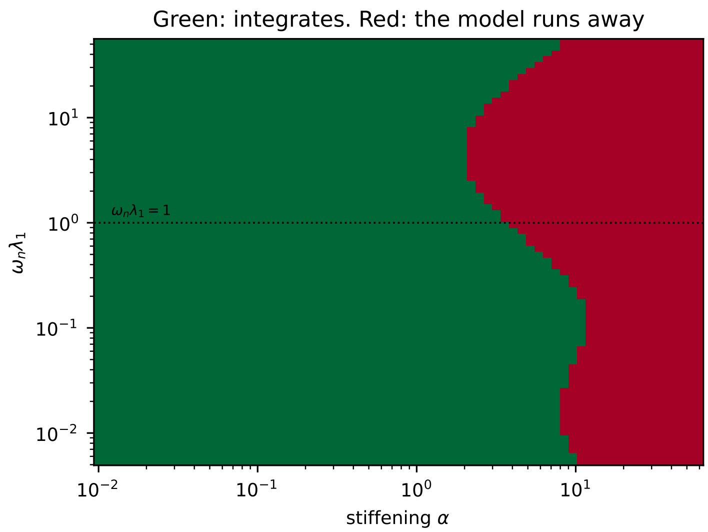

Figure 5: Where the winner can be integrated, at the fitted modulus and viscosity. Failure is governed principally by the stiffening, and the threshold falls to its lowest where the relaxation time is comparable to the bubble's own period — but it is a boundary that moves, not an isolated band.

model — a gap of 101.6 against run-to-run noise near 0.4, so the comparison is resolved by a wide margin. The tensor grid cannot say this on its own: it returns a single number with no uncertainty.

The same grid cannot estimate parameters. Refining it from 12 to 20 points per axis moves the log evidence by 17.3 and moves the best-fit point from $g = 658$, $\alpha = 1.52$ to $g = 144$, $\alpha = 8.86$, because the best cell is simply whichever node lands nearest the ridge. Refinement cannot fix this: the posterior standard deviations are 0.0018, 0.0025, 0.0055 and 0.078 in unit coordinates against a grid spacing of 0.053, so the posterior occupies about 10^{-7} of the prior volume and a uniform grid would need of order 10^{10} points. Discretisation error of 17.3 sits well inside the 101.6 gap, so the ranking is safe; the reported best-fit parameters are not. Figure 6 shows which models lose points.

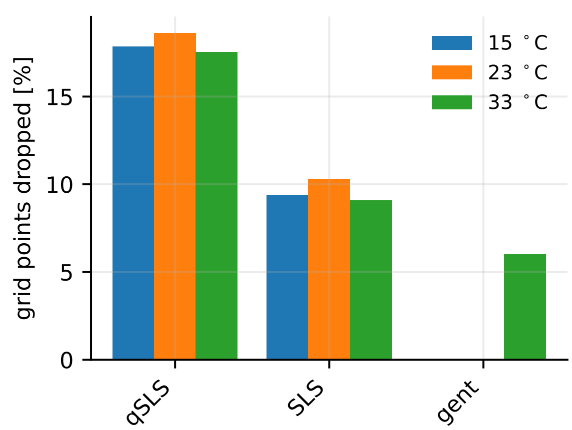

Figure 6: Fraction of each model's parameter grid that could not be integrated. Models not shown lost no points.

6 What the records determine, and what they do not

The winner fits, but not every parameter it contains is measured by these data.

Why the trade exists

Write $\hat{g} = G/p_\infty = 1/\mathrm{Ca}$ for the nondimensional modulus and $\lambda = R_{\mathrm{eq}}/R$ for the stretch the stress law sees. The equilibrium stress the Maxwell arm relaxes toward is

$$Z_e = R^3\left[\tfrac{3\alpha-1}{2}\,\hat{g}\,P(\lambda) + 2\alpha\,\hat{g}\,Q(\lambda)\right], \qquad \begin{aligned} P(\lambda) &= 5 - \lambda^4 - 4\lambda, \\ Q(\lambda) &= \tfrac{27}{40} + \tfrac{1}{8}\lambda^8 + \tfrac{1}{5}\lambda^5 + \lambda^2 - \tfrac{2}{\lambda}. \end{aligned} \tag{11}$$

Collecting powers of α separates it into two terms with different parameter dependence,

$$Z_e = \hat{g}\,\alpha\,A(\lambda) + \hat{g}\,B(\lambda), \qquad A = R^3\left[\tfrac{3}{2}P + 2Q\right], \qquad B = -\tfrac{1}{2}R^3 P. \tag{12}$$

This is exact; it reproduces the implemented expression to 2.5×10^{-14} over random states. Only the first term carries α, so $\hat{g}\alpha$ and $\hat{g}$ are separately determined only while *both* terms contribute.

They do not, at depth. As λ grows, $Q \to \frac{1}{8}\lambda^8$ dominates every other term in either bracket, so $A \to \frac{1}{4}R^3\lambda^8$ while $B \sim \frac{1}{2}R^3\lambda^4$, and

$$Z_e \;\longrightarrow\; \tfrac{1}{4}\,\hat{g}\,\alpha\,R^3\lambda^8, \tag{13}$$

a function of the *product* $\hat{g}\alpha$ alone. The likelihood cannot then distinguish a soft material that stiffens sharply from a stiff one that barely stiffens. Holding Z_e fixed in (12) gives the trade in closed form,

$$\frac{\partial \log \hat{g}}{\partial \log \alpha} = -\frac{\alpha A}{\alpha A + B} \;\longrightarrow\; -1 \quad (\lambda \to \infty), \tag{14}$$

and the share of the signal that separates the two parameters is

$$\frac{|\hat{g}B|}{|\hat{g}\alpha A|} = \frac{|B|}{\alpha|A|} \simeq \frac{2}{\alpha\lambda^4}. \tag{15}$$

Equations (14) and (15) are what the measurements see. These records collapse to $\lambda = 2.215$, where (14) gives -0.981 against -0.99 from the Fisher curvature — a prediction from the constitutive law, not a fit. The profile likelihood returns -0.80 because it averages over a range of λ, and the exponent is -0.861 at $\lambda = 1.5$. Separability falls as λ^{-4}: it is 2% of the signal at $\lambda = 2.2$ and 16% at $\lambda = 1.5$.

What the Fisher information measures

For Gaussian observations with known deviations the log-likelihood is $-\frac{1}{2}\sum_i (y_i - m_i(\theta))^2/\sigma_i^2$ up to a constant, so with the whitened sensitivity $J_{ij} = \sigma_i^{-1}\partial m_i/\partial\theta_j$ the Fisher information is exactly $F = J^\mathsf{T} J$, and $\Delta\chi^2 \simeq \Delta\theta^\mathsf{T} F \Delta\theta$. Taking θ to be the logarithms of the parameters makes the eigenvectors read as power-law combinations: along an eigenvector v the ratio v_g/v_α is the exponent in $g \propto \alpha^{v_g/v_\alpha}$. This is also the observability Gramian of the system augmented with $\dot{\theta} = 0$, which is why identifiability and observability are the same statement here.

One caution, since it caught us. When the sensitivities are taken with respect to a unit cube $u_j = (\log\theta_j - a_j)/w_j$ rather than $\log\theta_j$ directly, the eigenvector components carry the box widths: $\Delta\log\theta_j = w_j\,\tilde{v}_j$, so the exponent is $(w_g\tilde{v}_g)/(w_\alpha\tilde{v}_\alpha)$, not $\tilde{v}_g/\tilde{v}_\alpha$. With w of three and four decades the two differ by 25%.

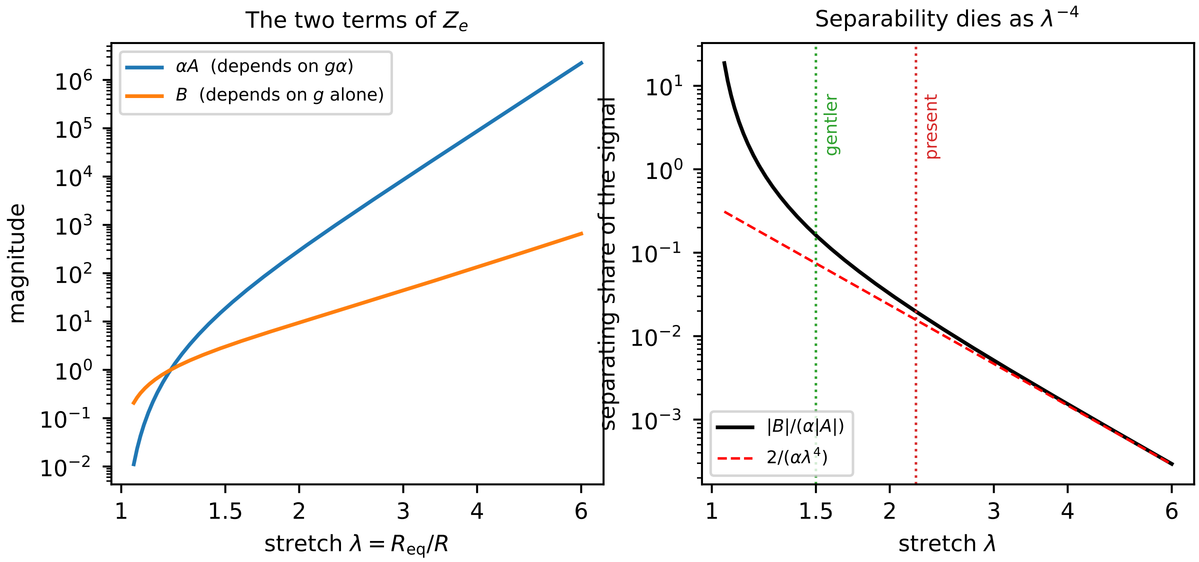

Figure 7: The two terms of (12) against collapse depth. Left: αA, which depends on the product $g\alpha$, outruns B, which carries g alone. Right: their ratio, the share of the signal that can separate the two parameters, following $2/(\alpha\lambda^4)$ once the collapse is deep. It is 2% at the depth these records reach and 16% at $\lambda = 1.5$.

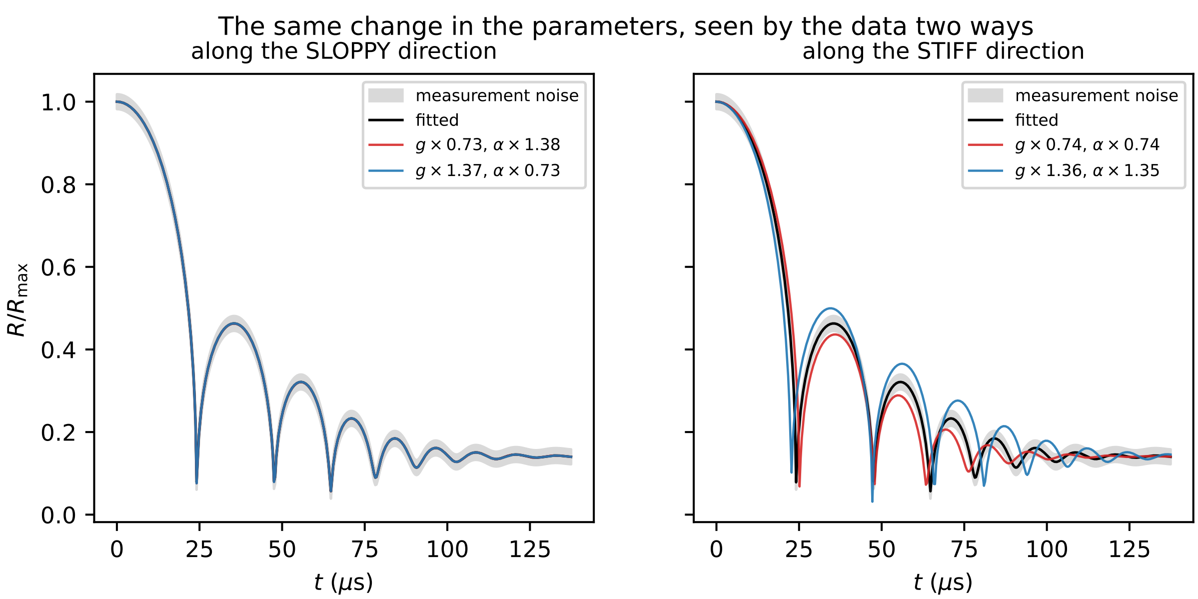

Figure 8: What an eigenvector means, shown rather than defined. The parameters are moved the same distance in log-space along each eigenvector. Along the sloppy direction a 27% change in the modulus, compensated by 38% in the stiffening, leaves the trajectory inside the measurement noise; along the stiff direction a change of the same size is obvious. The data cannot see the first and cannot miss the second.

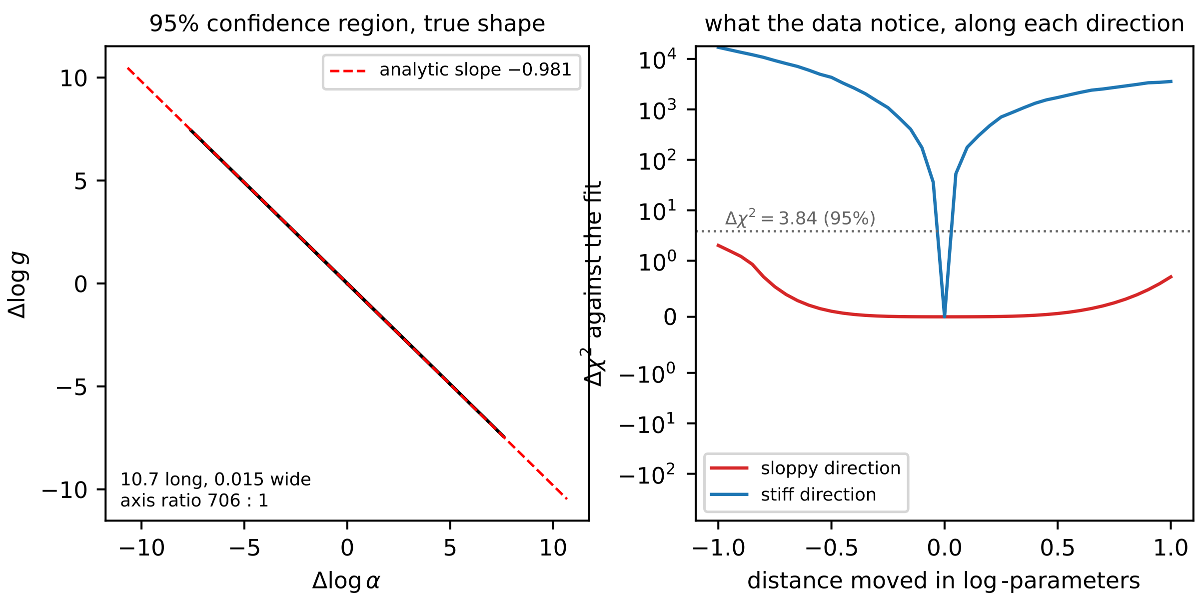

Figure 9: Left: the 95% region from the inverse Fisher information, 10.7 long and 0.015 wide in log-parameters — an axis ratio of 706, which is why it draws as a line, and the analytic slope of -0.981 lies along it. Right: the misfit measured by re-solving at perturbed parameters, not the quadratic approximation. Moving along the stiff direction costs 10^4 in $\Delta\chi^2$; moving the same distance along the sloppy one stays below the 95% threshold, so the parameters may shift by a factor of $e \approx 2.7$ without the data objecting.

Designing without assuming the model

Everything above optimises the Fisher information of one model at its fitted parameters, which is circular if that model is wrong. Two checks that do not privilege it.

The record is information-rich; the parameterisation is not. Drawing 400 trajectories from the posterior and taking the singular values of the centred ensemble, eleven modes rise above the trial noise floor of $0.018\,R_{\max}$. That is not a count of identifiable combinations — four varied parameters span at most a four dimensional manifold, and the extra modes are its curvature — but it does say the measurement can distinguish far more shapes than the fit extracts. The limit is how qSLS parameterises the physics, not the information in the record.

Discrimination does not privilege a model. T-optimality (Atkinson and Fedorov, *Biometrika* **62**, 1975) asks a different question: which design makes the candidates disagree most? The criterion is

$$T(d) \;=\; \min_{\theta_2} \left\| m_1(\hat{\theta}_1, d) - m_2(\theta_2, d) \right\| / \sigma, \tag{16}$$

and the minimisation is the whole content of it. A rival that can retune its parameters to imitate has not been discriminated against, so the separation must be measured against the rival's *best imitation*, not against its own fitted values.

Two distinct errors were made here, and both inflate T for the same structural reason: T is a *minimum*, so anything that stops short of it — evaluating at fixed parameters, or a minimisation that terminates early — reports discrimination that is not there. An earlier version of this table evaluated each rival at its own fit, giving 2.69 for qSLS–SLS at the present design. Re-fitting the rival with a single Nelder–Mead pass gave 0.67. Restarting and polishing that minimisation changed other entries again: at $1200\,\mu$m the reported 4.92 fell to 2.38, and a 0.1% change of design moved one loose evaluation from 7.25 to 1.36, which is the signature of the optimiser rather than the physics. The converged values are:

design	qSLS–SLS	qSLS–qKV	qSLS–NHKV
$277\,\mu$m, stretch 7.09 (present)	0.67	4.53	4.37
$277\,\mu$m, stretch 5.00	0.72	4.61	4.61
$277\,\mu$m, stretch 3.00	0.74	1.72	1.72
$199\,\mu$m, stretch 3.55 (E-optimal)	0.56	2.59	2.59
$895\,\mu$m, stretch 4.93	**2.39**	7.13	**7.13**
$910\,\mu$m, stretch 5.19	1.36	6.97	6.97
$1200\,\mu$m, stretch 7.09	2.38	5.03	5.63
$100\,\mu$m, stretch 20.0	1.26	**8.04**	4.17
$57\,\mu$m, stretch 19.5	1.64	5.53	5.00

At the present design SLS imitates qSLS to within 0.67 noise units, comfortably inside the measurement scatter. That figure is corroborated from an unrelated direction: a systematic offset of 0.668σ over 201 samples contributes $201 \times 0.668^2 = 89.7$ to χ^2, against the measured log-evidence gap of 101.6 between the two models. Two computations sharing no machinery, agreeing to about 12%.

These values are upper bounds, and the criterion is the reason. T is defined by an inner *global* minimisation, and that minimisation is multimodal. Nelder–Mead and Gauss–Newton both converge, to different basins: at 895 μm Nelder–Mead returns 2.39 where Gauss–Newton returns 1.52, and at 1200 μm the ordering reverses. Multistart Gauss–Newton — which analytic Jacobians make 44× cheaper than the derivative-free version, 34 s against 25 minutes for the whole table — finds a lower value in 12 of 27 cells, and a stationarity check built from the same Jacobians reports that 11 cells are still not at a minimum. Every improvement to the optimiser lowers the scores, because a better-optimised rival imitates better. The tabulated values should therefore be read as upper bounds on discrimination, and the true separations are smaller than any of them.

This is not a defect of the implementation but of the criterion. A minimum is attained at a point, so landing in the wrong basin gives a wrong answer with no indication that anything went wrong, and enumerating the outer design space exactly does not repair it: a dense grid over an unreliable integrand produces a front that is precisely wrong. Any criterion built on an argmin — T-optimality, profile likelihood, minimax design — inherits this.

The structural fix is to replace the minimum with an integral. Treating the model label as a discrete parameter makes discrimination an expected-information-gain problem over the augmented parameter (θ, m), whose inner quantity is the marginal likelihood $\int p(Y \mid \theta, M, d)\, \pi(\theta \mid M)\, \mathrm{d}\theta$. An integral accumulates over all modes rather than requiring that the best one be located, so multimodality stops being a failure mode and becomes a property of the integrand; a missed mode biases the evidence slightly instead of corrupting the criterion. It also unifies discrimination with the parameter-precision criteria above in one framework. The cost is that the marginal likelihood is prior-sensitive in a way T is not, so the prior becomes part of the criterion rather than a nuisance.

The criteria do not agree. Computed correctly, the criteria want different experiments. E-optimality peaks at 199 μm and stretch 3.55; separating qSLS from SLS or from NHKV is best at 895 μm and stretch 4.93; separating qSLS from qKV is best at 100 μm and stretch 20, near the opposite corner of the design space. At the E-optimal design the discrimination scores are the worst in the table, so the conflict is not marginal.

The reason is physical rather than numerical. Distinguishing a material with a relaxation mode from one without requires a collapse fast enough to excite that mode, whereas separating g from α requires a gentle collapse, since the term of Z_e carrying that distinction dies as λ^{-4}. The design cannot be both fast and gentle.

There is accordingly no single best design, and a scalar design objective is answering a question that has no scalar answer. What replaces it is the *Pareto front*: the set of designs on which no other design is better on every criterion at once. Enumerating a 14×18 grid over $(R_{\max}, \lambda)$ and applying that test leaves 12 designs undominated, spanning 50–1200 μm and stretch 4–18. Reporting that set, rather than a single optimum, is the honest summary of an experiment that must serve several questions at once; `pyimr.pareto.explore_tradeoff` computes it. Given the caution above, the T columns entering the domination test are upper bounds, so the front should be regarded as provisional in its discrimination coordinates while the E-optimality coordinate — an eigenvalue of $J^{\mathsf{T}} J$, with no inner optimisation — is not subject to that reservation.

Discrimination as an integral rather than a minimum

Everything above scores a design by T, which is defined through a *global minimum* over the rival's parameters. That definition is the source of the trouble in the previous subsection, and the trouble is structural rather than computational. This subsection derives a criterion with the minimum replaced by an integral, states precisely why that is more robust, and reports what it gives.

The model label is a parameter. Write $M \in \{M_1, M_2\}$ for the model index with prior $p(M)$, and let each model carry its own parameter θ with prior $\pi(\theta \mid M)$. Nothing distinguishes M from any other unknown, so the information a design d yields about it is the mutual information

$$I(M; Y \mid d) = \mathbb{E}_M\, \mathbb{E}_{Y|M,d}\left[\log \frac{p(Y \mid M, d)}{p(Y \mid d)}\right], \qquad p(Y \mid d) = \sum_M p(M)\, p(Y \mid M, d), \tag{17}$$

which is the expected reduction in entropy of M, exactly as the expected information gain used for the design study above is the expected reduction in entropy of θ. Discrimination and estimation are therefore the same criterion applied to different components of the augmented unknown (θ, M); they are not competing philosophies, and the earlier tension between T- and E-optimality is a tension between *which component we care about*, not between two schools.

The quantity entering (17) is the marginal likelihood, or evidence,

$$p(Y \mid M, d) = \int p(Y \mid \theta, M, d)\, \pi(\theta \mid M)\, \mathrm{d}\theta, \tag{18}$$

the same object as (4) but now at a hypothetical design and for hypothetical data. It is an integral over θ, and that single fact is what the rest of this subsection turns on.

The criterion actually computed. Rather than (17) we score a design by the expected log Bayes factor in favour of the model that generated the data,

$$U(d) = \mathbb{E}_{Y \sim p(\cdot|M_1,d)}\left[\log p(Y \mid M_1, d) - \log p(Y \mid M_2, d)\right] = D_{\mathrm{KL}}(p(\cdot \mid M_1, d) \,\|\, p(\cdot \mid M_2, d))\,. \tag{19}$$

Recognising it as a Kullback–Leibler divergence between the two prior-predictive distributions settles its properties without further work. By Gibbs' inequality $U(d) \geq 0$, with equality if and only if the two predictives agree almost everywhere: *a model cannot discriminate against itself, at any design*. That is a sharp, falsifiable statement about the estimator, and it is the property the implementation is tested against — two independently drawn banks from one family must score zero, and any systematic excess is bias. Related to (17) by $I(M; Y \mid d) = \frac{1}{2}[D_{\mathrm{KL}}(p_1\|\bar{p}) + D_{\mathrm{KL}}(p_2\|\bar{p})]$ with $\bar{p}$ the equal mixture, so (19) is the same comparison read against the rival rather than against the mixture, and is the more direct statement of "how strongly will this experiment favour the truth".

Why an integral is robust where a minimum is not. Both criteria require an inner computation that can go wrong. The asymmetry is in how the error propagates.

For T, any optimiser returns $\hat{T} = \|m_1 - m_2(\hat{\theta}_2)\|/\sigma$ at whatever point it stopped, and since T is a minimum, $\hat{T} \geq T$ always. If $\hat{\theta}_2$ lies in the correct basin but is imprecise, stationarity gives $\hat{T} - T = O(\|\hat{\theta}_2 - \theta_2^\star\|^2)$ and the error is negligible. If it lies in the *wrong basin*, the error is $O(1)$: a fixed, unbounded overstatement with nothing in the returned value to indicate it. This is what was measured — Nelder–Mead and Gauss–Newton returning 2.39 and 1.52 at one design and reversing at another.

For the evidence, suppose the posterior has modes $k = 1, \ldots, K$ contributing $Z = \sum_k Z_k$, and a procedure finds only a subset $\mathcal{S}$. Then

$$\log \hat{Z} - \log Z = \log\left(1 - \sum_{k \notin \mathcal{S}} \frac{Z_k}{Z}\right), \tag{20}$$

so the error is controlled by the *posterior weight* of what was missed. Missing a subdominant mode costs $O(Z_k/Z)$ and missing a negligible one costs nothing. Because the contributions add, a multistart that finds several modes can sum them rather than choose between them, and each additional mode found monotonically improves the estimate. There is no analogue for a minimum, where finding more candidates only ever changes which single one is reported.

This is the whole argument for the change, and it is worth stating plainly: multimodality is a *failure mode* for a minimum and merely a *property of the integrand* for an integral.

Estimating the evidence I: prior sampling, and why it fails here. The direct estimator draws $\theta_s \sim \pi(\theta \mid M)$ and averages the likelihood,

$$\hat{Z} = \frac{1}{S}\sum_{s=1}^{S} p(Y \mid \theta_s, M, d), \qquad w_s = p(Y \mid \theta_s, M, d), \tag{21}$$

which is unbiased for Z, though $\log \hat{Z}$ is biased low by Jensen's inequality. Its reliability is carried by the spread of the weights, summarised by the effective sample size

$$S_{\text{eff}} = \frac{\left(\sum_s w_s\right)^2}{\sum_s w_s^2} \;\in\; [1, S], \tag{22}$$

which equals S when all draws contribute equally and 1 when a single draw dominates. It costs nothing to compute and it is the only thing standing between this estimator and a confidently reported number with no content.

That it must fail here can be seen in advance. With N samples the log-likelihood scales as $\log p(Y \mid \theta) \simeq -\frac{1}{2}N\bar{d}(\theta)$ for a mean squared discrepancy $\bar{d}$, so weights are $w_s \propto \exp[-\frac{1}{2}N\bar{d}(\theta_s)]$ and the ratio of the two largest is exponential in N. Useful draws are those falling inside the posterior, whose width is $O(N^{-1/2})$ in each of p directions, so the expected number of them is

$$\mathbb{E}[\#\text{useful}] \;\sim\; S \cdot \frac{\text{vol(posterior)}}{\text{vol(prior)}} \;\sim\; S\,N^{-p/2}. \tag{23}$$

For the present study $N = 201$, $p = 4$ and $S = 220$, giving $220 \times 201^{-2} \approx 5 \times 10^{-3}$: fewer than one useful draw. The measurement agrees exactly — $S_{\text{eff}} = 1.0$ at *every* design tried, with reported values between 793 and 10 979 nats that are pure noise. The standard errors were 56 to 130 throughout and looked entirely respectable. Without (22) the failure is invisible.

Estimating the evidence II: Laplace. Expanding $\log[p(Y \mid \theta)\pi(\theta)]$ to second order about its maximum $\hat{\theta}$ and performing the resulting Gaussian integral gives, as in (5),

$$\log Z \simeq \log p(Y \mid \hat{\theta}) + \log \pi(\hat{\theta}) + \tfrac{p}{2}\log 2\pi - \tfrac{1}{2}\log\det H, \qquad H = -\nabla^2 \log p(Y \mid \theta)\big|_{\hat{\theta}}. \tag{24}$$

For Gaussian data $H = J^{\mathsf{T}}J + \sum_i r_i \nabla^2 r_i$, and dropping the second term — the Gauss–Newton approximation, valid when residuals are small or the model nearly linear — leaves $H \simeq J^{\mathsf{T}}J$, already computed by the fit. Parameterising so the prior is uniform on the unit cube makes $\pi(\hat{\theta}) = 1$, and with whitened residuals $r = (m - y)/\sigma$,

$$\log Z \simeq -\tfrac{1}{2}\|r\|^2 - \tfrac{N}{2}\log\left(2\pi\sigma^2\right) + \tfrac{p}{2}\log 2\pi - \tfrac{1}{2}\log\det\left(J^{\mathsf{T}}J\right). \tag{25}$$

The first two terms are the best fit; the last two are the Occam factor, and their sign is what prevents a more flexible model from winning on fit alone. Crucially for the present purpose, (25) does not care how sharp the likelihood is, which is precisely the regime that destroyed (21). It is affordable only because the fit uses analytic Jacobians: derivative-free minimisation of the same objective costs 55 s against 7.6 s, and the Jacobian needed for $\det(J^{\mathsf{T}}J)$ is a by-product of the solve rather than an extra cost.

The two criteria are consistent where both can be computed. The criteria are not independent, and the relation between them is a check on both. If the true model fits to within noise, $\chi_1^2 \approx N$, while a rival that cannot fit leaves a residual of root-mean-square $T\sigma$, so $\chi_2^2 \approx NT^2$. Substituting into the leading term of (25),

$$\log \mathrm{BF} \;\approx\; \tfrac{1}{2}\left(\chi_2^2 - \chi_1^2\right) \;\approx\; \tfrac{N}{2}\left(T^2 - 1\right) \xrightarrow{T \gg 1} \tfrac{N}{2}T^2, \tag{26}$$

up to the Occam terms, which T has no way to express. Testing (26) against the measured values:

design	T (vs SLS)	$\frac{N}{2}T^2$ predicted	measured U (nats)
199 μm, stretch 3.55	0.56	31.5	35.4 ± 12.2
895 μm, stretch 4.93	1.52	232	150 ± 73

Two computations sharing no machinery — one a constrained least-squares minimum, the other a marginal likelihood by Laplace — agreeing to within their stated uncertainty. The residual gap at 895 μm is of the right sign and size for the Occam penalty that qSLS pays for its fourth parameter.

What it costs, and where it is not yet trustworthy. Two prices are paid for removing the minimum, and both are real.

Prior sensitivity. Z depends on $\pi(\theta \mid M)$ through the normalisation of (18), and widening a prior by a factor c in each of p directions divides the evidence by roughly c^p without changing the fit at all. This is the Lindley–Jeffreys effect, and it is not a defect to be engineered away: a model whose prior is diffuse really is less supported by data that pin it down. But it means the prior is part of the criterion and must be stated, where T needed no prior. Centring each rival's prior on its fit to the *existing* record produced a criterion that measured prior placement instead of discriminability, with every Laplace expansion invalid; the results reported here use the wide, physically motivated box of the model-selection grid, identical across models on every shared parameter.

Boundary pinning. Equation (24) presumes an interior maximum. In 50–90% of the fits at each design, the rival's best parameters lie on a face of the prior box, where the expansion does not apply and the reported value is not to be trusted. That this happens is itself informative — a rival that can imitate qSLS only by driving a parameter to the edge of physical plausibility has, in a meaningful sense, been discriminated against — but expressing that requires a criterion built for a constrained maximum rather than a quadratic expansion about an interior one. Determining which parameter pins, and whether the pinning is physical or an artefact of the box, is the outstanding step.

The honest summary is that the integral criterion is better founded than T and not yet more reliable on this problem. The multimodality that made T untrustworthy is genuinely gone; what replaces it is a dependence on the prior, now explicit rather than hidden, and a boundary condition that the current estimator does not handle.

Why the design criteria disagree

The classical criteria are all functionals of the same F: $\det F$ (D), $\operatorname{tr} F^{-1}$ (A), $\lambda_{\min}$ (E), and $c^\mathsf{T} F^{-1} c$ for one combination (c). A design change that merely amplifies the signal — a larger bubble, a longer record, less noise — multiplies F by a scalar s. Then $\lambda_{\min} \to s\lambda_{\min}$ and $\det F \to s^p \det F$, so D and E both improve, while

$$\kappa = \lambda_{\max}/\lambda_{\min} \tag{27}$$

is unchanged. Degeneracy is a property of the *directions*, and only a design that alters them can fix it. That is why the E-optimal search preferred a larger bubble (895 μm, $\lambda_{\min} = 12.8$) while the conditioning and the eigenvector composition preferred a gentler collapse at the present size (277 μm at stretch 5, $\kappa = 4.2 \times 10^3$, and the modulus-stiffening share of the worst direction falling from 87% to 66%). Both are correct answers to different questions, and (15) says which question matters here.

Modulus and stiffening trade against each other. Two independent analyses agree. The Fisher information at the fitted optimum has an eigenvalue spectrum spanning six decades, and the eigenvector of its smallest eigenvalue is almost purely a shear-modulus-against-stiffening direction, with weights -0.796 and $+0.604$ and contributions of -0.003 and -0.032 from viscosity and relaxation time. Separately, profiling the likelihood in α — fixing it and re-optimising everything else — gives $g \propto \alpha^{-0.80}$ across two decades. Sampling the posterior gives a third, independent measurement: its widest direction is $(-0.682, +0.002, -0.082, +0.727)$ in the logarithms of $(g, \mu, \lambda_1, \alpha)$, and $\log g$ is 98.7% anticorrelated with $\log \alpha$.

The three exponents are -0.99 from local curvature, -0.94 from the posterior covariance, and -0.80 from the profile. They agree that the trade is near-reciprocal and disagree in the third digit, which is expected rather than troubling: the valley is curved, so a curvature estimate at one point and a chord measured across two decades are not measuring the same slope. The correlation is the more robust statement.

So viscosity and relaxation time are cleanly determined, while modulus and stiffening are constrained only in combination. The identifiable quantity is closer to $g\,\alpha^{0.8}$ than to either alone, and a fitted α should be reported with its correlation to g rather than as an independent material property.

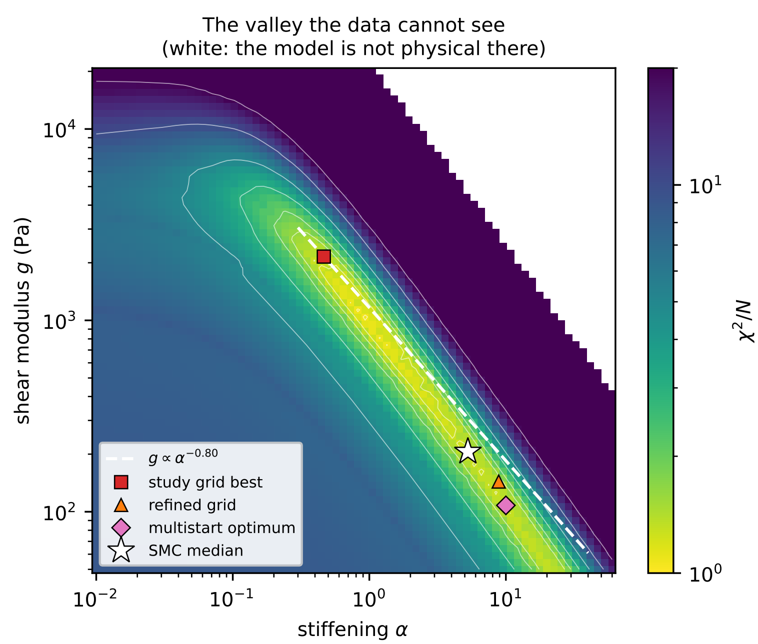

Figure 10: χ^2/N over the modulus–stiffening plane, at the fitted viscosity and relaxation time. The dashed line is $g \propto \alpha^{-0.80}$, predicted from the Fisher curvature at a single point; it lies along the floor of the valley. Four "best fits" from four methods are strung out along that floor rather than disagreeing with one another. The white region at upper right is where the model is not physical.

Figure 10 shows why the four estimates in this study disagree about g by an order of magnitude while agreeing about the fit: they are different points on the floor of a single flat-bottomed trench.

The stiffening is nonetheless identified, once the prior allows it. Sampling with the ceiling on α placed at 1, 10 and 100:

ceiling	log evidence	median α	97.5%
1	2162.9	0.85	0.97
10	2173.9	6.37	9.04
100	2172.4	5.27	8.73

Widening the prior tenfold moves the median only from 6.37 to 5.27 and the evidence by 1.5 out of 2173, so α is identified near 5 with 97.5% of posterior mass below about 9. A prior-dominated parameter would have tracked its bound. The ceiling of 1 shows what genuine truncation looks like: the median pins against it and 11 log units of evidence are lost.

This corrects a natural reading of the profile likelihood alone, which appears flat to the boundary only because the boundary sits where the posterior's upper tail does.

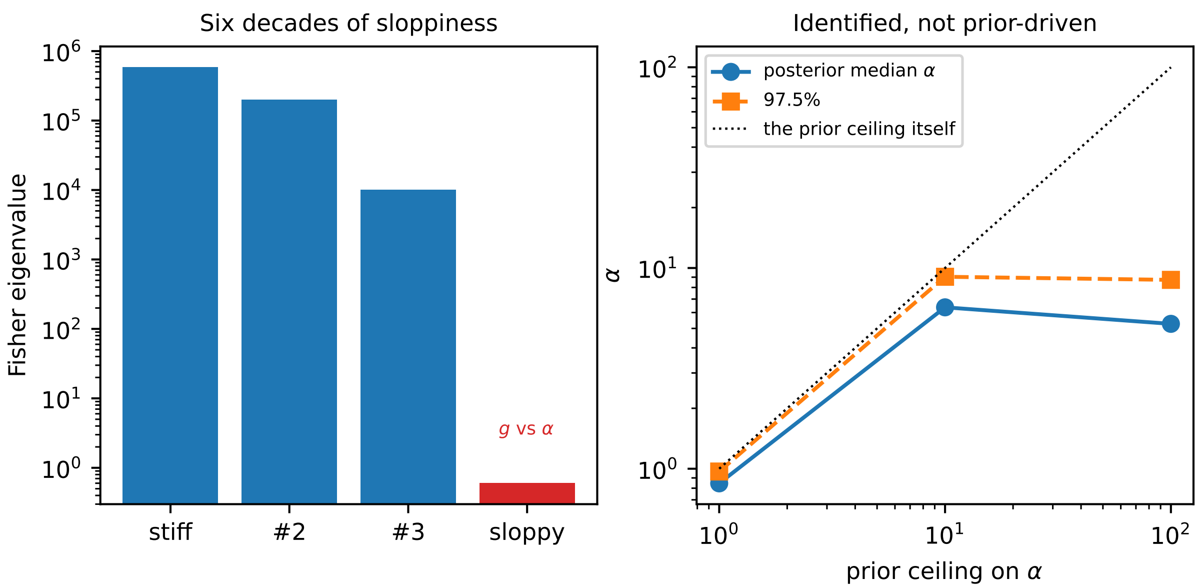

Figure 11: Left: the Fisher spectrum at the optimum, six decades between the stiffest and sloppiest directions, with the sloppiest being modulus against stiffening. Right: the posterior for α against the ceiling imposed on its prior. It tracks the ceiling while truncated and departs from it once the ceiling is loose, which is what distinguishes an identified parameter from a prior-driven one.

Consequences for the numbers reported here. Three methods that resolve the posterior — sampling, a refined grid, and continuous optimisation — agree that g lies near 10^2 Pa. The coarse grid used for the ranking reports 2154 Pa. The ranking does not depend on this, but any material property quoted from the grid does. We also note that a modulus of that size is implausibly soft for gelatin, which is the degeneracy above making itself felt: the fit buys a lower residual by trading modulus against stiffening, along a direction the data barely constrain.

One candidate was never genuinely exercised. The Gent solid appears in the comparison but cannot be tested on these records. Its energy diverges as $I_1 - 3 \to J_m$, and a Gent solid soft enough to be distinguishable from Neo-Hookean predicts a collapse it cannot sustain, so it locks up. Where it does integrate, it differs from Neo-Hookean by 1.7×10^{-4} of $R_{\max}$ against measurement noise near 2×10^{-2} — two orders of magnitude below what the data can resolve. Any statement about strain-stiffening laws here excludes it.

The noise model sees only repeatability. Using trial-to-trial spread as the noise means $\chi^2/N \approx 1$ certifies agreement to within experimental repeatability. A systematic error common to every trial would not appear in that spread, and would show up instead as exactly the kind of residual excess we report — 20 to 35% above the noise floor. This study cannot separate that from genuine model inadequacy.

A Statistical background

This appendix develops, from first principles, the results the main text uses. It is self-contained and assumes only calculus and elementary probability.

A.1 Inference as conditioning

Write θ for parameters, y for data, M for a model. The product rule $p(\theta, y) = p(y \mid \theta)\, p(\theta) = p(\theta \mid y)\, p(y)$ rearranges to Bayes' theorem,

$$p(\theta \mid y, M) = \frac{p(y \mid \theta, M)\, \pi(\theta \mid M)}{p(y \mid M)}, \qquad p(y \mid M) = \int p(y \mid \theta, M)\, \pi(\theta \mid M)\, \mathrm{d}\theta. \tag{28}$$

The denominator is a normalising constant in θ, which is why it is ignored when estimating parameters and why it is the entire content of model comparison: applying (28) at the level of models gives

$$\frac{p(M_1 \mid y)}{p(M_2 \mid y)} = \underbrace{\frac{p(y \mid M_1)}{p(y \mid M_2)}}_{\text{Bayes factor } B_{12}} \cdot \frac{p(M_1)}{p(M_2)}. \tag{29}$$

Everything the data say about which model to prefer is in B_{12}. Jeffreys' conventional reading takes $\log B > 1$ as substantial and $\log B > 5$ as decisive; the gap reported here is over 100, which is why its numerical resolution had to be checked rather than assumed.

A.2 The score, and two definitions of information

Let $\ell(\theta) = \log p(y \mid \theta)$. The *score* is $u = \nabla_\theta \ell$. Since $\int p(y \mid \theta)\, \mathrm{d}y = 1$ for every θ, differentiating under the integral gives

$$\mathbb{E}_\theta[u] = \int \frac{\nabla p}{p}\, p\, \mathrm{d}y = \nabla \int p\, \mathrm{d}y = 0, \tag{30}$$

so the score has zero mean at the true parameter. Its covariance defines the Fisher information, $F = \mathbb{E}[uu^\mathsf{T}]$. Differentiating the same identity twice gives the equivalent and more useful form

$$F = \mathbb{E}\left[-\nabla^2 \ell\right], \tag{31}$$

the expected curvature. The two agree because $\nabla^2 \log p = \nabla^2 p/p - (\nabla p/p)(\nabla p/p)^\mathsf{T}$ and the first term integrates to zero. Curvature and variance-of-score are the same object, which is the reason a sharply peaked likelihood and a precisely determined parameter are the same statement.

A.3 The Cramér–Rao bound

For an unbiased estimator $\hat{\theta}(y)$, differentiate $\mathbb{E}[\hat{\theta}] = \theta$ under the integral to get $\mathbb{E}[\hat{\theta}\, u^{\mathsf{T}}] = I$. Applying the Cauchy–Schwarz inequality to the pair $(\hat{\theta} - \theta,\ u)$ in the scalar case gives $\mathrm{Var}(\hat{\theta})\, \mathbb{E}[u^2] \geq 1$, i.e. $\mathrm{Var}(\hat{\theta}) \geq 1/F$; the multivariate statement is

$$\mathrm{Cov}(\hat{\theta}) \succeq F^{-1}. \tag{32}$$

So F^{-1} is a floor on achievable precision, attained asymptotically by maximum likelihood: $\sqrt{N}(\hat{\theta}_N - \theta) \to \mathcal{N}(0, F_1^{-1})$ under regularity. This is the licence for reading F^{-1} as a covariance and its eigenvectors as the axes of a confidence ellipsoid — and the regularity conditions are exactly what near-singular F threatens, which is why sloppiness is worth detecting rather than tolerating.

A.4 Gaussian data, and why $F = J^{\mathsf{T}} J$

With independent Gaussian observations of known deviation, $\ell = -\frac{1}{2}\sum_i (y_i - m_i(\theta))^2/\sigma_i^2 + \text{const}$. Then

$$\frac{\partial \ell}{\partial \theta_j} = \sum_i \frac{y_i - m_i}{\sigma_i^2}\frac{\partial m_i}{\partial \theta_j}, \qquad -\frac{\partial^2 \ell}{\partial \theta_j \partial \theta_k} = \sum_i \frac{1}{\sigma_i^2}\left[\frac{\partial m_i}{\partial \theta_j}\frac{\partial m_i}{\partial \theta_k} - (y_i - m_i)\frac{\partial^2 m_i}{\partial \theta_j \partial \theta_k}\right]. \tag{33}$$

Taking the expectation kills the second bracket because $\mathbb{E}[y_i - m_i] = 0$, leaving $F = J^{\mathsf{T}} J$ with $J_{ij} = \sigma_i^{-1} \partial m_i/\partial \theta_j$. Note what was discarded: the curvature of the model. At the optimum the residuals are small and the Gauss–Newton form is accurate; far from it, or with a badly misspecified model, F and the observed Hessian differ, and the discrepancy is itself a diagnostic.

A.5 The Laplace approximation and the Occam factor

To evaluate $Z = \int e^{\ell(\theta)}\pi(\theta)\, \mathrm{d}\theta$, expand about the maximum $\hat{\theta}$ where $\nabla \ell = 0$:

$$\ell(\hat{\theta} + \delta) = \ell(\hat{\theta}) - \tfrac{1}{2}\delta^{\mathsf{T}} H \delta + O(\|\delta\|^3), \qquad H = -\nabla^2 \ell(\hat{\theta}). \tag{34}$$

Holding π constant over the peak and using $\int e^{-\delta^{\mathsf{T}} H\delta/2} \mathrm{d}\delta = (2\pi)^{p/2}|H|^{-1/2}$,

$$Z \simeq e^{\ell(\hat{\theta})}\, \pi(\hat{\theta})\, (2\pi)^{p/2}\, |H|^{-1/2}. \tag{35}$$

The correction is $O(N^{-1})$ relative for regular problems. Writing the posterior volume as $V_{\text{post}} = (2\pi)^{p/2}|H|^{-1/2}$ and the prior density as $1/V_{\text{prior}}$ where it is flat, the second factor is $V_{\text{post}}/V_{\text{prior}}$: the fraction of the space the data leave standing. A model with more parameters has a larger V_{prior} and pays for it unless the fit improves enough to compensate. Nothing was added by hand; the penalty is the Jacobian of the integral.

Taking logarithms of (35), using $H \approx N F_1$ for N independent observations, and dropping terms that do not grow with N,

$$-2 \log Z \simeq -2\ell(\hat{\theta}) + p \log N + O(1), \tag{36}$$

which is the Bayesian information criterion. BIC is therefore not an alternative to the evidence but a coarse asymptotic of it, valid when the posterior is Gaussian, the parameters identified, and N large. AIC answers a different question — expected out-of-sample Kullback–Leibler divergence — and is not an approximation to Z at all. Where a model is sloppy, H is near-singular, $|H|^{-1/2}$ is enormous, and the asymptotic form fails while the integral remains well defined.

A.6 Sloppy models

Write the eigen-decomposition $F = V\Lambda V^{\mathsf{T}}$ with $\lambda_1 \geq \cdots \geq \lambda_p$. The misfit near the optimum is $\Delta\chi^2 = \sum_k \lambda_k c_k^2$ for $\delta = \sum_k c_k v_k$, so moving a distance c along v_k costs $\lambda_k c^2$ and the 95% contour lies at $c_k = \sqrt{3.84/\lambda_k}$. Multiparameter models of this kind generically show spectra spanning many decades, with a few stiff directions and many sloppy ones — the observation of Gutenkunst, Sethna and co-workers, and the reason parameter values in such models are often far less determined than their predictions.

The distinction to keep is between a rank-deficient F, where a direction is *structurally* unidentifiable and no experiment helps, and a merely ill-conditioned F, where the direction is determined but weakly. In logarithmic parameters, an eigenvector is a power-law combination, and the invariant it leaves fixed is read off its components.

A.7 Correlated errors

Let r be the residual vector with covariance C. The Gaussian log-likelihood is

$$-2\ell = r^{\mathsf{T}} C^{-1} r + \log|C| + N \log 2\pi, \tag{37}$$

and with the Cholesky factorisation $C = LL^{\mathsf{T}}$ this becomes $\|L^{-1}r\|^2 + 2\sum_k \log L_{kk} + \text{const}$: whitening turns generalised least squares back into ordinary least squares on transformed residuals. Ignoring the off-diagonal terms, i.e. using $\operatorname{diag}(C)$, does not merely lose efficiency — it misstates the amount of information. For a stationary sequence with autocorrelation ρ_k,

$$\operatorname{Var}\left(\sum_{i=1}^{N} r_i\right) = N\sigma^2\left(1 + 2\sum_{k=1}^{N-1}\left(1 - \tfrac{k}{N}\right)\rho_k\right) \xrightarrow{N\to\infty} N\sigma^2\left(1 + 2\sum_{k\geq 1}\rho_k\right), \tag{38}$$

so the sequence carries the information of $N_{\text{eff}} = N/(1 + 2\sum \rho_k)$ independent draws. The bracket is the spectral density of the residual sequence at zero frequency, which is the cleaner way to see why a long-range correlation is so costly.

Correlated residuals in a *fitted* model usually indicate model discrepancy rather than correlated measurement noise. Kennedy and O'Hagan formalise this by writing $y = m(\theta) + \delta(\cdot) + \varepsilon$ with δ a Gaussian process, and Brynjarsdóttir and O'Hagan show that omitting δ leaves parameter estimates biased *and* overconfident, while an unconstrained δ absorbs the physics. Neither failure is visible in χ^2.

A.8 Optimal experimental design

A design ξ determines what is measured and hence $F(\xi, \theta)$. Since F is a matrix and designs must be ranked, a scalar functional is required, and the classical choices are

$$\Phi_D = \log\det F, \quad \Phi_A = -\operatorname{tr} F^{-1}, \quad \Phi_E = \lambda_{\min}(F), \quad \Phi_c = -c^{\mathsf{T}} F^{-1} c, \tag{39}$$

each maximised. In Bayesian terms, Φ_D is the expected Kullback–Leibler gain from prior to posterior under a Gaussian approximation, which is why it is the default: it maximises information about *all* parameters jointly. That default is wrong whenever one direction is the question, because $\log\det F = \sum_k \log\lambda_k$ is dominated by the large eigenvalues.

Because F depends on θ, a local design presupposes an answer. The standard repairs are to average over a prior, $\Phi(\xi) = \mathbb{E}_\theta[\Phi(F(\xi,\theta))]$, or to take the worst case over a plausible set. Sequential design uses the posterior from completed experiments as the prior for the next, which is what is done here.

When the model itself is uncertain, all of the above presuppose the wrong thing. T-optimality (Atkinson and Fedorov) maximises instead the lack of fit of the inferior model,

$$\Phi_T(\xi) = \min_{\theta_2} \left\| m_1(\hat{\theta}_1;\xi) - m_2(\theta_2;\xi) \right\|^2, \tag{40}$$

which asks which experiment the candidates most disagree about. Its virtue here is that it shares no assumptions with the Fisher criteria, so agreement between them is evidence rather than restatement.

B Reproducing

```
.venv/bin/python examples/measured_selection.py <dataset> [thermal Nt] [workers]
.venv/bin/python docs/writeup/collect.py 16     # records results.json
.venv/bin/python docs/writeup/figures.py        # draws every figure from it
```

Every number and figure in this document comes from `results.json`; none is transcribed by hand.